AI STARTUPS, VISUALIZED · BRAIN

Physical Intelligence, explained.

Can one AI model learn to control different robots? Physical Intelligence (PI) is trying to build exactly that. Here is the story the reel told, step by step, in the words of PI's own research posts.

EVERY QUOTE IS FROM PHYSICAL INTELLIGENCE'S BLOG, CHECKED 18 OCT 2026. POSTS CHANGE, SO RE-CHECK BEFORE RELYING ON A DETAIL. FUNDING AND VALUATION ARE DELIBERATELY NOT COVERED.

The problem

Most robots are specialists. PI writes that "most commercial robots operate in tightly controlled environments like factories or warehouses" (S2), and that unlike the web, "there is no such treasure trove of robot data" (S1).

The idea, in four steps

01 π0 · one model, many robots

Trained on data from eight different robots plus open-source robot datasets and internet-scale vision-language pre-training. It turns camera images and an instruction into motor commands.

> "π0 uses Internet-scale vision-language pre-pretraining, open-source robot manipulation datasets, and our own datasets consisting of dexterous tasks from 8 distinct robots" · "output motor commands at a high frequency, up to 50 times per second"

PI BLOG · S1 · OCT 2024

02 π0.5 · says the next step, then moves

It first writes a short “high level” step in words (e.g. “pick up the pillow” for “clean the bedroom”), then picks motor commands. It was tested in homes that were not in its training data.

> “we first ask it to output a ”high level” action expressed in text, and then ask it to follow this high level action by choosing an appropriate robot motor command” · “All experiments were done in homes that were not in the training data.”

PI BLOG · S2 · APR 2025

03 π0.7 · a task learned on one body, done on another

It folded laundry on a heavy two-arm UR5e system with no folding data collected on that robot. The folding data came from a much smaller two-arm robot.

> “We tasked π0.7 to control this robot to fold laundry, even though we did not collect any data of laundry folding with this robot, and to our surprise it could do this consistently.”

PI BLOG · S3 · APR 2026

04 API beta · fine-tune for your own robot

Companies can upload their data, fine-tune on PI-managed compute and run inference through a hosted gateway. It is a beta.

> “You upload data, launch fine-tuning on compute and infrastructure we manage, and access fine-tuned models via a hosted gateway to run inference.”

PI BLOG · S4 · OCT 2026

The honest part

In PI’s own words

“π0.5 is far from perfect, and it often makes mistakes both in terms of its high-level semantic deductions and motor commands.” (S2)

“It does not always succeed on the first try” (S2). The home results are from PI’s experiments, not a product you can buy, and the API is a beta.

Five questions to ask any robot demo

- [ ] Which robots and tasks is the demo on, and was that robot in the training data for that task?
- [ ] Is the footage autonomous, and at what playback speed?
- [ ] How often does it fail, and how does it recover?
- [ ] What data would you need to fine-tune it for your robot?
- [ ] What does “beta” cover today (access, support, limits)?

Glossary

VLA (vision-language-action model)

A model that takes images and text in and produces robot actions out.

Cross-embodiment

Learning from many different robot bodies, and doing a task on a body it was not trained on for that task.

Motor commands

The low-level signals that move a robot’s joints and gripper, many times per second.

Fine-tuning

Extra training on your own data so a general model fits your robot and task.

Sources

S1 · π0: Our First Generalist Policy (Oct 31, 2024)
S2 · π0.5: a VLA with Open-World Generalization (Apr 22, 2025)
S3 · π0.7: a Steerable Model with Emergent Capabilities (Apr 16, 2026)
S4 · Frontier Models for Every Robot: Announcing our API beta (Oct 13, 2026)
S5 · Physical Intelligence homepage

The reel’s product imagery (robot illustrations, evaluation clips, API poster) is official Physical Intelligence media from pi.website, credited ”Product imagery: Physical Intelligence (official)”. No press kit is published.

Independent concept, not affiliated with or endorsed by Physical Intelligence. Made with code and AI tools; AI voice.

VYNEMA STUDIO · AI Startups, Visualized